

The cube of a sum. The cube of a difference

Four terms, coefficients 1, 3, 3, 1, and the signs that alternate in the second formula.

LESSON 102–103

2 × 40 MIN

MATEMATIKA 7, §46

BEYOND STAGE 8

LESSON CLOCK

15'

20'

25'

20'

5'

From the square to the cube	15 min
The two formulae	20 min
Expanding with coefficients	25 min
Reading a cube backwards	20 min
Homework	5 min

LEARNING OBJECTIVES

- Derive $(a + b)^3$ from $(a + b)^2$ by one more multiplication.
- Write both cube formulae from memory, with the right signs.
- Expand cubes of binomials with coefficients and two letters.
- Recognise a perfect cube and cube numbers mentally.

TEXTBOOK

National	pp. 297–303
Cambridge	Stage 8 extension
Workbook	Extension sheet 2A

01

Explanation

From the square to the cube

$(a + b)^3$ is $(a + b)^2$ multiplied by one more bracket — nothing more mysterious than that.

STEP	WORKING
the square	$a^2 + 2ab + b^2$
times a	$a^3 + 2a^2b + ab^2$
times b	$a^2b + 2ab^2 + b^3$
collect	$a^3 + 3a^2b + 3ab^2 + b^3$

$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

THE COEFFICIENTS ARE 1, 3, 3, 1

The square had 1, 2, 1 and the cube has 1, 3, 3, 1. Each row of this triangle of numbers is built from the one above it — a pattern you will meet again in Grade 11 under the name of the binomial theorem.

The two formulae

$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

$(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$

TERM	IN $(A + B)^3$	IN $(A - B)^3$
a^3	+	+
$3a^2b$	+	−
$3ab^2$	+	+
b^3	+	−

IN THE DIFFERENCE THE SIGNS ALTERNATE, AND THE LAST ONE IS MINUS

Unlike $(a - b)^2$, which ends in $+ b^2$, the cube ends in $- b^3$ — because an odd power of a negative number stays negative. Ending a cube with a plus is the standard slip.

Expanding with coefficients

EXPRESSION	A	B	EXPANSION
$(x + 2)^3$	x	2	$x^3 + 6x^2 + 12x + 8$
$(x - 1)^3$	x	1	$x^3 - 3x^2 + 3x - 1$
$(2x + 1)^3$	2x	1	$8x^3 + 12x^2 + 6x + 1$
$(a - 2b)^3$	a	2b	$a^3 - 6a^2b + 12ab^2 - 8b^3$

THE WHOLE TERM IS CUBED, COEFFICIENT INCLUDED

In $(2x + 1)^3$ the first term is $(2x)^3 = 8x^3$, not $2x^3$. Every power in the expansion applies to the bracketed term as a whole.

Reading a cube backwards

Four terms with coefficients in the ratio $1 : 3 : 3 : 1$ and matching cube roots make a perfect cube.

EXPRESSION	AS A CUBE
$x^3 + 6x^2 + 12x + 8$	$(x + 2)^3$
$x^3 - 3x^2 + 3x - 1$	$(x - 1)^3$
$8x^3 + 12x^2 + 6x + 1$	$(2x + 1)^3$
$x^3 + 3x^2 + 3x + 2$	not a cube — the last term should be 1

CUBING NUMBERS IN THE HEAD

$21^3 = (20 + 1)^3 = 8000 + 1200 + 60 + 1 = 9261$, and $19^3 = (20 - 1)^3 = 8000 - 1200 + 60 - 1 = 6859$. The alternating signs are easiest to see in exactly these calculations.

02 Worked examples

EXAMPLE 1 Expand $(2x + 1)^3$.

① $a = 2x, b = 1.$

② $a^3 = 8x^3$

The coefficient is cubed too.

③ $3a^2b = 3 \cdot 4x^2 \cdot 1 = 12x^2$

④ $3ab^2 = 3 \cdot 2x \cdot 1 = 6x$, and $b^3 = 1.$

ANSWER

$$8x^3 + 12x^2 + 6x + 1$$

EXAMPLE 2 Expand $(a - 2b)^3$.

① a^3

② $-3a^2(2b) = -6a^2b$

③ $+3a(2b)^2 = 12ab^2$

④ $-(2b)^3 = -8b^3$

ANSWER

$$a^3 - 6a^2b + 12ab^2 - 8b^3$$

EXAMPLE 3 Find 19^3 mentally.

$$① \quad 19 = 20 - 1$$

$$② \quad 20^3 = 8000$$

$$③ \quad -3 \cdot 400 \cdot 1 = -1200 \text{ and } +3 \cdot 20 \cdot 1 = +60.$$

$$④ \quad 8000 - 1200 + 60 - 1 = 6859$$

ANSWER

6859

03 Interactive model

Build the row 1, 3, 3, 1 from 1, 2, 1 on the board by adding neighbours; the class then predicts 1, 4, 6, 4, 1 without being told, and the pattern is theirs.

Signs and coefficients of a cube

The coefficients of $(a + b)^3$ are:

1, 2, 1

1, 3, 3, 1

1, 1, 1, 1

1, 4, 6, 4, 1

How many terms does a cube of a binomial have?

2

3

4

6

The last term of $(a - b)^3$ is:

$+ b^3$

$- b^3$

$+ b^2$

$- b^2$

The last term of $(a - b)^2$ is:

The first term of $(2x + 1)^3$ is:

$(x - 1)^3$ equals:

19^3 equals:

The row after 1, 3, 3, 1 is:

1, 4, 4, 1

1, 4, 6, 4, 1

1, 3, 4, 3, 1

1, 5, 10, 5, 1

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Cube of a sum	Yig‘indining kubi	Куб суммы
Cube of a difference	Ayirmaning kubi	Куб разности
Cube	Kub	Куб
Coefficient	Koeffitsiyent	Коэффициент
Alternating signs	Almashinuvchi ishoralar	Чередующиеся знаки
Perfect cube	To‘la kub	Полный куб
Expansion	Yoyilma	Разложение
Pattern	Qonuniyat	Закономерность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The coefficients in $(a + b)^3$ are:

1, 2, 1

1, 3, 3, 1

1, 1, 1, 1

1, 4, 6, 4, 1

$(a - b)^3$ ends in:

$+ b^3$

$- b^3$

$+ b^2$

$- b^2$

$(x - 1)^3$ equals:

$x^3 - 1$

$x^3 - 3x^2 + 3x - 1$

$x^3 + 3x^2 + 3x + 1$

$x^3 - 3x + 1$

The first term of $(2x + 1)^3$ is:

The middle terms of $(x + 2)^3$ are:

21^3 equals:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $(x + 1)^3$

ANSWER $x^3 + 3x^2 + 3x + 1$

2. $(x - 1)^3$

ANSWER $x^3 - 3x^2 + 3x - 1$

3. $(x + 2)^3$

ANSWER $x^3 + 6x^2 + 12x + 8$

4. $(x - 2)^3$

ANSWER $x^3 - 6x^2 + 12x - 8$

5. $(a + 3)^3$

ANSWER $a^3 + 9a^2 + 27a + 27$

6. The coefficients of a cube

ANSWER $1, 3, 3, 1$

7. 11^3

ANSWER 1331

Medium · the standard question

7 PROBLEMS

1. $(2x + 1)^3$

ANSWER $8x^3 + 12x^2 + 6x + 1$

2. $(3a - 1)^3$

ANSWER $27a^3 - 27a^2 + 9a - 1$

3. $(a - 2b)^3$

ANSWER $a^3 - 6a^2b + 12ab^2 - 8b^3$

4. $x^3 + 6x^2 + 12x + 8$ as a cube

ANSWER $(x + 2)^3$

5. $8x^3 + 12x^2 + 6x + 1$ as a cube

ANSWER $(2x + 1)^3$

6. 21^3

ANSWER 9261

7. 19^3

ANSWER 6859

Hard · stretch and reason

7 PROBLEMS

1. $(x + \frac{1}{x})^3$

ANSWER $x^3 + 3x + \frac{3}{x} + \frac{1}{x^3}$

2. $(a + b)^3 - (a - b)^3$

ANSWER $6a^2b + 2b^3$

3. $(a + b)^3 + (a - b)^3$

ANSWER $2a^3 + 6ab^2$

4. If $a + b = 5$ and $ab = 6$, find $a^3 + b^3$

ANSWER 35

5. $(2a - 3b)^3$

ANSWER $8a^3 - 36a^2b + 54ab^2 - 27b^3$

6. Is $x^3 + 3x^2 + 3x + 2$ a perfect cube?

ANSWER No — the last term should be 1

7. The next row after 1, 3, 3, 1

ANSWER 1, 4, 6, 4, 1

07 Homework

Homework — 5 tasks

Write the four signs before you write any numbers.

1. Expand $(x + 4)^3$ and $(x - 3)^3$.

2. Expand $(3x + 2)^3$.

3. Expand $(2a - b)^3$.

4. Write $x^3 - 9x^2 + 27x - 27$ as a cube.

5. Find 31^3 and 29^3 by the formulae.